Telco Customer Churn Prediction

Predicting which telecom customers are likely to leave, using **Logistic Regression** and **Random Forest**. HyperionDev Machine Learning capstone project.

Overview

- **Data:** IBM Telco Customer Churn dataset (7,043 customers, 21 columns, about 27% churn)
- **Tools:** Python, pandas, Matplotlib, seaborn, scikit-learn, Jupyter
- **Steps:** data cleaning, dummy encoding, EDA, min-max scaling, 75/25 train-test split, model training and evaluation

What I did

- Cleaned the data: converted TotalCharges to numeric, removed 11 rows with missing values and dropped customerID.
- Encoded Churn as 1/0 and turned categorical features into dummy variables, using drop_first to avoid the dummy variable trap.
- Explored the data with a correlation chart, a tenure histogram, a charges scatter plot and a tenure-by-churn box plot.
- Scaled all features to 0–1 with min-max scaling and split the data 75% train / 25% test.
- Trained a Logistic Regression model and a tuned Random Forest (2,000 trees, sqrt features, 50 max leaf nodes, bootstrapping, OOB scoring).
- Evaluated both models with accuracy, OOB error, confusion matrices, precision and recall, then recommended the best model for the business.

Key insights

- Customers with fibre optic internet, electronic check payments and high monthly charges churn most.
- Long tenure and two-year contracts are linked to customers staying.
- Most churn happens in the first year.

Results

Metric	Logistic Regression	Random Forest
Accuracy	0.791	0.794
Precision	0.619	0.648
Recall	0.518	0.454

The Random Forest OOB score was 0.81, close to its test accuracy, so it generalises well.

Conclusion

Missing a customer who leaves costs more than offering an unnecessary discount, so recall matters most. **Logistic Regression is the better model.** It catches 29 more churners and is easier to explain.

Run it

```
pip install pandas numpy matplotlib seaborn scikit-learn jupyter
jupyter notebook telco_customer_churn.ipynb
```